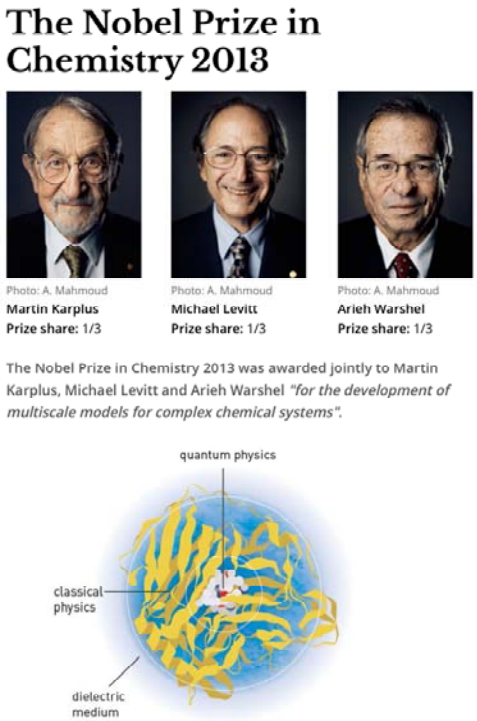
통계역학

제1장 분자동역학 모의실험의 원리

중요 학습 내용

- **Newtonian equation of motions**
- **Force fields**
- **Time integration**
- **Initial distributions**
- **Periodic boundary conditions and minimum image conventions**

계산화학의 중요성

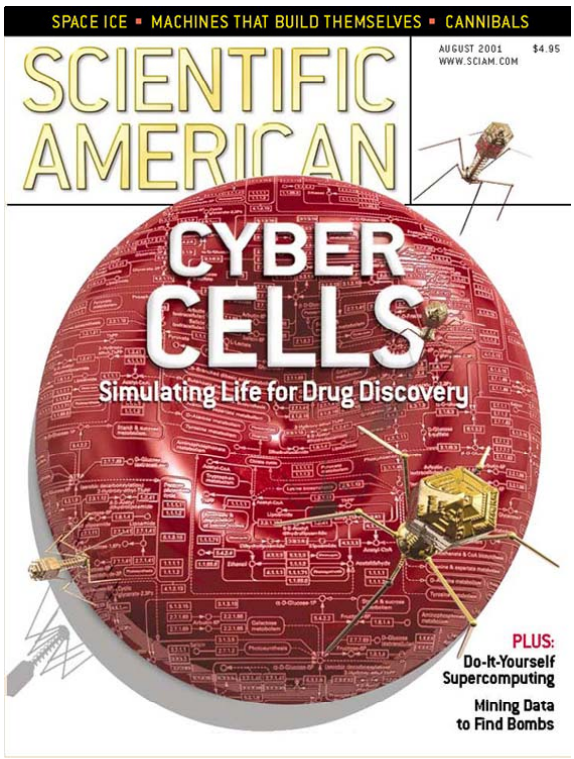




재료과학과 계산화학



생명과학과 계산화학



"Computational Chemistry confirms that new antimicrobial kills *E.Coli* within 30 seconds", *Science Daily, 2016*



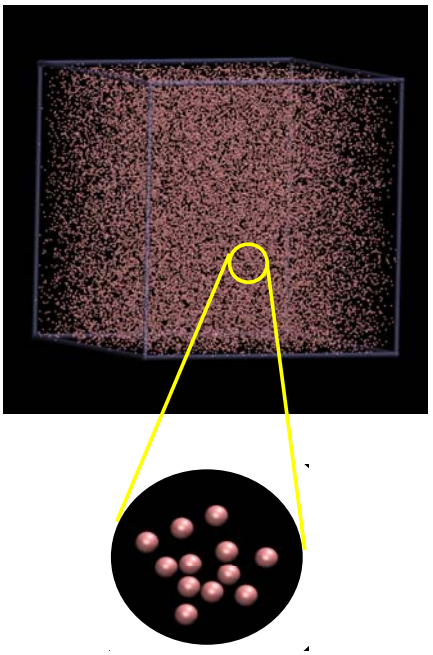
Newtonian equation of motion

- Total energy of the system

$$E = \sum_{i=1}^{N} \frac{1}{2} m_i v_i^2 + V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$

- Newtonian equation of motion

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = -\nabla V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = F(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$



$$V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = V^{\text{bonded}}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) + V^{\text{non-bonded}}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$

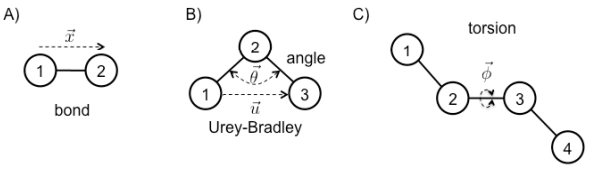
Potential energy for bonded interactions

- Potential energy for bonded interaction

$$V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = V^{\text{stretching}} + V^{\text{bending}} + V^{\text{torsion}} + V^{\text{improper}} + \cdots$$

- Stretching interaction energy

$$V^{\text{stretch}} = \sum_i k_{i,\text{stretch}} (r_i - r_{i,0})^2$$



- Bending interaction energy

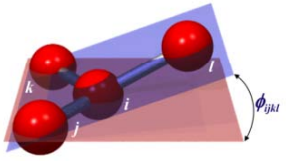
$$V^{\text{bend}} = \sum_i k_{i,\text{bend}} (\theta_i - \theta_{i,0})^2 + \sum_j k_{\text{UB}} (u_i - u_{i,0})^2$$

- Torsional interaction energy

$$V^{\text{torsion}} = \sum_i V_{i,\text{torsion}} \left[1 + \cos(n\phi_i - \phi_{i,0})\right]$$

- Improper interaction

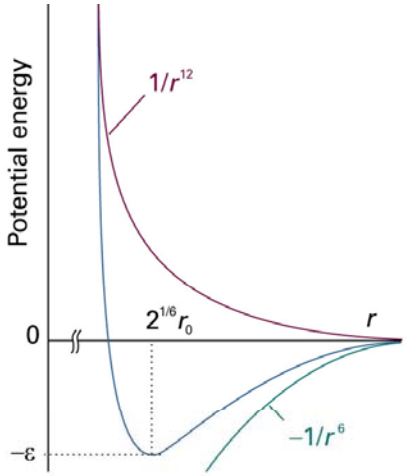
$$V^{\text{improper}} = \sum_i k_{i,\text{improper}} (\phi_{ijkl} - \phi_{ijkl,0})^2$$



Potential energy for non-bonded interactions

- Lennard-Jones potential energy for a monoatomic molecule

$$V^{LJ}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N} \sum_{j>i}^{N} 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^{6} \right]$$



- Electrostatic potential energy

$$V^{el}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N} \sum_{j>i}^{N} \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}}$$

where

$$\frac{1}{r_{ij}} = \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

	ϵ (kcal/mol)	σ (Å)
He	0.02027	2.28
Ne	0.09340	2.72
Ar	0.2381	3.41
Kr	0.3259	3.83

Empirical force fields

1. CHARMM force fields
2. AMBER force fields
3. GROMOS force fields
4. OPLS force fields
5. MM force fields
6. And so on.....

Integrating Newtonian equations of motion

- Methods for numerical integrations of Newtonian equations of motion
  1) Verlet algorithm

$$\mathbf{r}_i(t+\delta t) \approx 2\mathbf{r}_i(t) - \mathbf{r}_i(t-\delta t) + \frac{\mathbf{F}_i(t)}{m_i}\delta t^2 + O(\delta t^4)$$

$$\mathbf{v}_i(t) \approx \frac{\mathbf{r}_i(t+\delta t) - \mathbf{r}_i(t-\delta t)}{2\delta t} + O(\delta t^2)$$

  2) Leapfrog form of Verlet algorithm

$$\mathbf{r}_i(t+\delta t) \approx \mathbf{r}_i(t) + \mathbf{r}_i(t+\frac{\delta t}{2})\delta t + O(\delta t^4)$$

$$\mathbf{v}_i(t+\frac{\delta t}{2}) \approx \mathbf{v}_i(t-\frac{\delta t}{2}) + \frac{\mathbf{F}_i(t)}{m_i}\delta t + O(\delta t^3)$$

  3) Velocity Verlet algorithm
  4) Gear's predictor-corrector algorithm

Integrating Newtonian equations of motion -Velocity Verlet algorithm (1)

- Taylor expansion

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f^{(3)}(x)}{3!}h^3 + \cdots$$

- Advancing the position of a particle

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \mathbf{r}_i'(t)\delta t + \frac{1}{2}\mathbf{r}_i''(t)\delta t^2 + \cdots$$

$$\approx \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\mathbf{a}_i(t)\delta t^2$$

$$\approx \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2$$

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2$$

Integrating Newtonian equations of motion -Velocity Verlet algorithm (2)

- Advancing the velocity of a particle

$$\mathbf{v}_i(t+\frac{1}{2}\delta t) = \mathbf{v}_i(t) + \frac{1}{2}\mathbf{v}_i'(t)\delta t + \frac{1}{4}\mathbf{v}_i''(t)\delta t^2 + \cdots \approx \mathbf{v}_i(t) + \frac{1}{2}\mathbf{a}(t)\delta t + \frac{1}{4}\mathbf{v}_i''(t)\delta t^2$$

$$\approx \mathbf{v}_i(t) + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t + \frac{1}{4}\mathbf{v}_i''(t)\delta t^2$$

$$\mathbf{v}_i(t+\frac{1}{2}\delta t) = \mathbf{v}_i(t+\delta t) - \frac{1}{2}t\mathbf{v}_i'(t+\delta t)\delta + \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2 + \cdots$$

$$\approx \mathbf{v}_i(t+\delta t) - \frac{1}{2}\mathbf{a}(t+\delta t)\delta t + \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2$$

$$\approx \mathbf{v}_i(t+\delta t) - \frac{1}{2}\frac{\mathbf{F}_i(t+\delta t)}{m_i}\delta t + \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2$$

$$\mathbf{v}_i(t+\delta t) \approx \mathbf{v}_i(t+\frac{1}{2}\delta t) + \frac{1}{2}\frac{\mathbf{F}_i(t+\delta t)}{m_i}\delta t - \frac{1}{4}\mathbf{v}_i''(t+\delta t)\delta t^2$$

$$\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \frac{1}{2m_i}\delta t\left(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\right) + O(\delta t^3)$$

Integrating Newtonian equations of motion -Velocity Verlet algorithm summary

- Advancing the position of a particle

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\frac{\mathbf{F}_i(t)}{m_i}\delta t^2$$

- Advancing the velocity of a particle

$$\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \frac{1}{2m_i}\delta t\left(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\right) + O(\delta t^3)$$

Application of velocity Verlet algorithm to a harmonic oscillator model

- Newtonian equation of motion for a harmonic oscillator

$$m_a \frac{d^2 x_a}{dt^2} = -k x_a \Rightarrow x_a(t) = A\cos(\omega t) + B\sin(\omega t) \qquad (\omega = \sqrt{k/m_a})$$

만약 $x_a(0) = 0,\ v_a(0) = v_0$ 라면,

$$\boxed{x_a(t) = \frac{v_0}{\omega}\sin(\omega t)}$$

- 예: $k = 4,\ m_a = 1,\ v_0 = 1 \Rightarrow \omega = 2$

$$x_a(t) = \frac{1}{2}\sin(2t), \qquad v_a(t) = \cos(2t)$$

$$\begin{aligned} x_a(t+\delta t) &= x_a(t) + v_a(t)\delta t + \frac{1}{2}\frac{F_a(t)}{m}\delta t^2 \\ &= x_a(t) + v_a(t)\delta t - 2x_a(t)\delta t^2 \end{aligned}$$

$$\begin{aligned} v_i(t+\delta t) &= v_i(t) + \frac{1}{2m_i}\big(F_i(t) + F_i(t+\delta t)\big)\delta t \\ &= v_i(t) - \big(2x_a(t) + 2x_a(t+\delta t)\big)\delta t \end{aligned}$$

Ensemble

- Ensemble: a collection or set of microstates
- Types of ensembles
  1) Microcanonical ensemble: describes an isolated system. N, V, and E fixed
  2) Canonical ensemble: represents a closed system. N,V, and T fixed
  3) Grand canonical ensemble: describes an open system. μ, V, and T fixed
  4) Isothermal-isobaric ensemble: represents a closed system having flexible walls. N, P, and T fixed
- Ensemble averge is equal to time average
- Molecular dynamics (MD) simulations take time average for a thermodynamic property
- Monte Carlo (MC) simulations take ensemble average for the same property.

Maxwell-Boltzmann distributions for initial conditions

- The Maxwell-Boltzmann distribution of molecular velocities in x direction

$$f(v_x)dv_x = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \exp\left[-\frac{mv_x^2}{2k_B T}\right] dv_x$$

- The Maxwell-Boltzmann distribution of molecular velocities in three dimensions

$$f(v_x)f(v_x)f(v_x)dv_x dv_y dv_z = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{m(v_x^2+v_y^2+v_z^2)}{2k_B T}\right] dv_x dv_y dv_z$$

$$= \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{mv^2}{2k_B T}\right] dv_x dv_y dv_z$$

Kinetic model of gases

- The total momentum change during Δt

$$\text{momentum change} = \text{impact} = \frac{NAv_x\Delta t}{2V} \times 2mv_x$$

$$= \frac{NAmv_x^2\Delta t}{V}$$

- Average pressure

$$P = \frac{\text{impact}}{A\Delta t} = \frac{Nm\langle v_x^2\rangle}{V}$$

- Average kinetic energy of a molecule in *x* direction

$$\text{average KE of a molecule} = \frac{1}{2}m\langle v_x^2\rangle = \frac{1}{2}\frac{PV}{N} = \frac{1}{2}\frac{Nk_B T}{N} = \frac{1}{2}k_B T$$

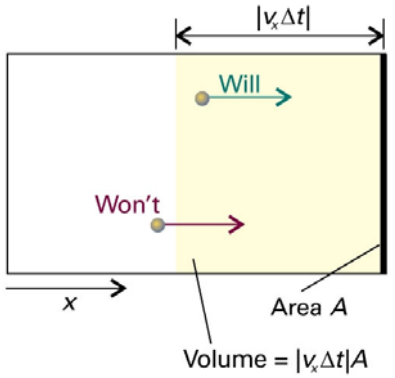


Fig. 21.2 A molecule will reach the wall on the right within an interval Δt if it is within a distance $v_x\Delta t$ of the wall and travelling to the right.

Velocity rescaling

- Average kinetic energy of a molecule

$$\text{average KE of a molecule} = \frac{1}{2}m\langle v^2\rangle = \frac{1}{2}m\langle v_x^2 + v_y^2 + v_z^2\rangle$$
$$= \frac{1}{2}m\langle v_x^2\rangle + \frac{1}{2}m\langle v_y^2\rangle + \frac{1}{2}m\langle v_z^2\rangle = \frac{1}{2}k_BT + \frac{1}{2}k_BT + \frac{1}{2}k_BT = \frac{3}{2}k_BT$$

- Scaling factor

$$\frac{1}{2}m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle = \frac{3}{2}(N-1)k_BT$$
$$\frac{1}{2}m\sum_{i=1}^{N}\langle (v_n^f)^2\rangle^f = \frac{3}{2}(N-1)k_BT_{ref}$$

$$\text{scaling factor} = \left(\frac{3(N-1)T_{ref}}{m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle}\right)^{1/2}$$

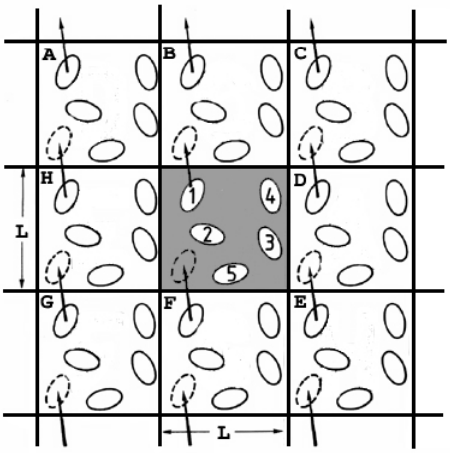
$$\sum_{n=1}^{N}\langle (v_n^f)^2\rangle = \frac{(N-1)T_{ref}}{(N-1)T}\sum_{n=1}^{N}\langle (v_n^i)^2\rangle$$
$$= \sum_{n=1}^{N}\left\langle \left\{\left(\frac{(N-1)T_{ref}}{(N-1)T}\right)^{1/2} v_n^i\right\}^2\right\rangle$$
$$= \sum_{i=1}^{N}\left\langle \left\{\left(\frac{3(N-1)T_{ref}}{m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle}\right)^{1/2} v_n^i\right\}^2\right\rangle$$

Temperature control methods

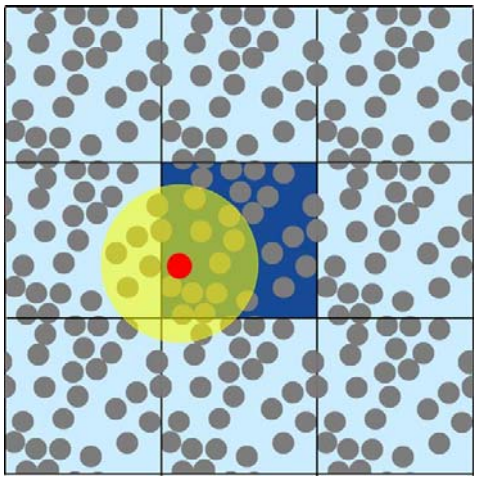
1. Velocity rescaling method
2. Nose-Hoover method
3. Temperature reassignment methods
4. Lowe-Anderson thermostat method
5. Langevin thermostat method

Periodic boundary conditions and minimum image convention

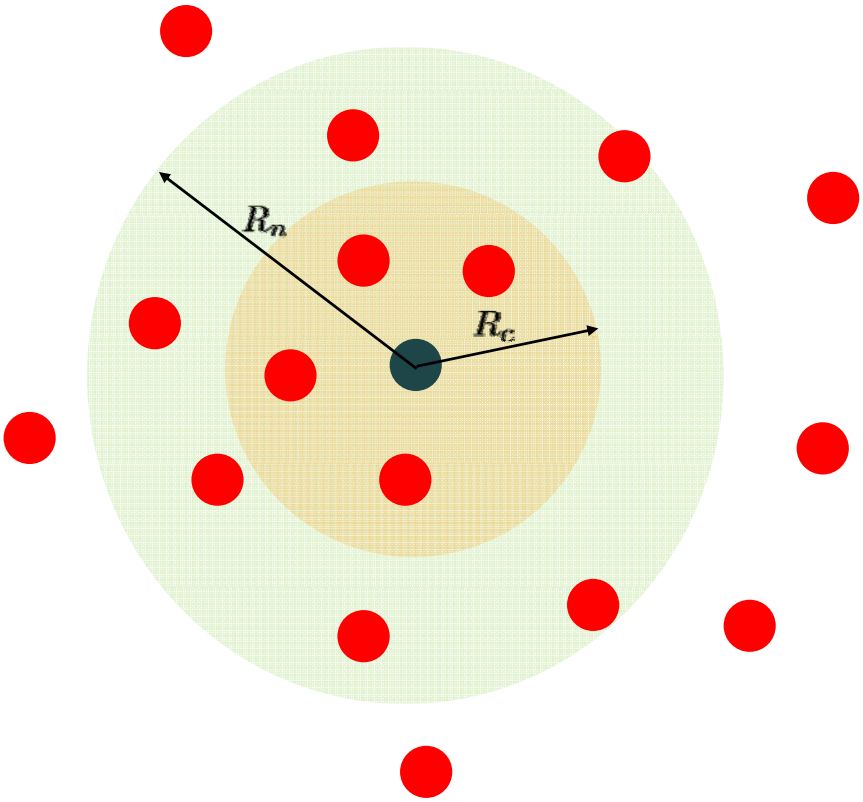
- Periodic boundary conditions (PBC)



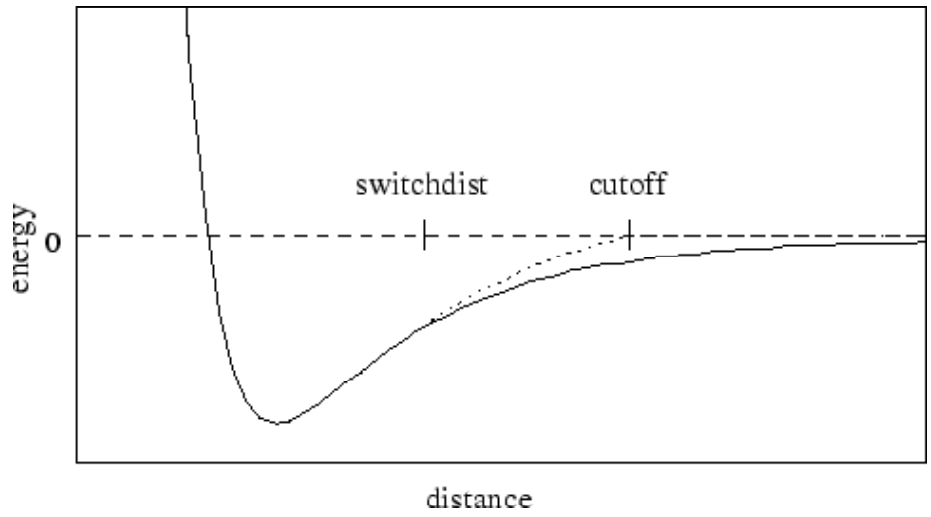
- Minimum Image Convention (MIC)



Cutoff distance and neighbor list



Switching function



Pressure calculation

- Virial theorem for pressure

$$P = \rho k_B T + \frac{1}{3V}\left\langle \sum_{i=1}^{N}\sum_{j>i}^{N} \mathbf{r}_{ij} \cdot \mathbf{F}_{ij} \right\rangle$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $\mathbf{F}_{ij}$ is the force acting on particle *i* by particle *j*.

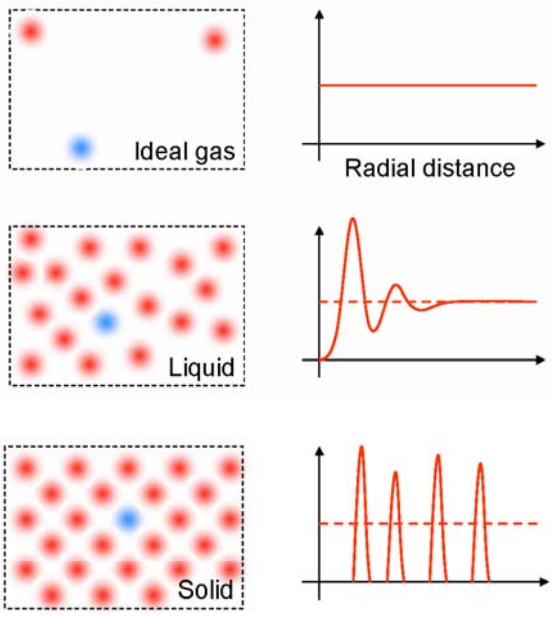
Root mean square displacement (RMSD)

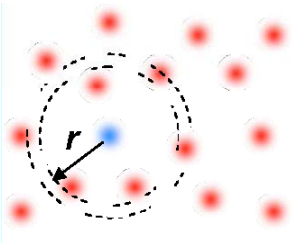
- Definition of RMSD

$$RMSD_{\text{protein}}(t_j) = \sqrt{\frac{\sum_{\alpha=1}^{N_\alpha}(\mathbf{r}_\alpha(t_j) - \mathbf{r}_\alpha(0))^2}{N_\alpha}}$$

Radial distribution function - structured fluids

- Radial distribution function of gas, liquid, and solid





- Calculate the number of particles inside a radial segement ***(r,r+Δr)*** from one specific particle in the system.
- Average the distribution over all particles in the system.
- Divide the (average) number of particles in a segment by the segment volume.
- Finally, normalize thus obtained radial density profile with the bulk density ***N/V*** of the system.